CYBERTRACE

Digital Forensics Timeline Analyzer

PS-013: Incident Timeline and Evidence Collection for Small Teams

From Scattered Logs to an Evidence-Backed Incident Timeline

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Problem Description

Small IT teams struggle to reconstruct security incidents because evidence is scattered, inconsistent and hard to verify.

- **Scattered evidence:** logs sit on different systems and use different formats.
- **Unreliable time:** time zones and clock differences make event order uncertain.
- **Hidden gaps:** investigators may not know which sources stopped reporting or which records failed to parse.
- **Slow manual work:** connecting events across systems by hand takes time and effort.
- **Unclear links:** how two events are related is often not explained.
- **Changing evidence:** collected evidence must be preserved, hashed and traceable.

Objective: build a lightweight, evidence-aware platform that turns scattered logs from authorized systems into a searchable, chronological incident timeline, while keeping original evidence intact and showing the limits of the data.

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What a small team needs to investigate an incident, and what we plan to build first [1]

Context

- Forensic guidance (NIST SP 800-86) describes four steps: collection, examination, analysis and reporting [1].
- Security logging guidance stresses that log records must be complete and protected [2].
- Users: small IT teams, cybersecurity students, incident responders and small organisations.
- Large monitoring and forensic platforms exist (see next slide). CyberTrace targets a lightweight, evidence-aware workflow for small teams.

Planned prototype scope

- Central investigation server
- Controlled Linux endpoint agent
- Sources: Linux authentication and web-server logs
- Secure log ingestion
- Original evidence kept, SHA-256 (hash) verified
- Timestamp normalization and uncertainty reporting
- Collection-coverage and parsing-failure tracking
- Rule-based cross-system event correlation
- Searchable timeline; PDF and CSV reports
- Repeatable synthetic incident scenarios

Status: proposal stage. Features are planned; no evaluation results have been measured yet.

AIORI 3 Literature & RFC Review

Standards guide our process; existing open-source tools are baselines for comparison.

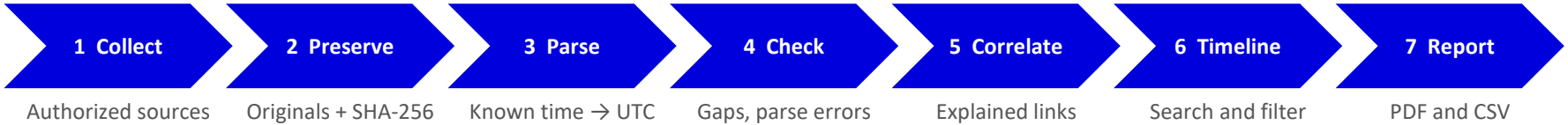
Source	What it provides (per its documentation)	Role in CyberTrace
NIST SP 800-86 [1]	Guide to integrating forensic techniques into incident response	Basis for the collect, examine, analyze, report workflow
OWASP Logging Cheat Sheet [2]	Guidance on security logging and log protection	Informs which log fields and protections matter
Wazuh [3]	Open-source security monitoring platform	Comparison baseline
Velociraptor [4]	Endpoint monitoring and digital-forensics collection	Comparison baseline
Timesketch [5]	Collaborative forensic timeline analysis	Comparison baseline
Plaso [6]	Timeline generation from forensic artifacts	Comparison baseline

No claim is made that these tools lack any capability. A feature-by-feature comparison, using the same synthetic incidents, is planned (see Deployment Timeline).

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Proposed Solution

CyberTrace turns scattered logs from authorized systems into one searchable incident timeline, with evidence limits kept visible.



How it helps

- Gives small teams one place to see events from several systems in time order.
- Shows what is uncertain or missing instead of hiding it.
- Not a replacement for a full SIEM (Security Information and Event Management) or forensic suite.

Planned technology stack

- **Python** – collection and analysis logic
- **FastAPI** – central ingestion APIs
- **React** – dashboard
- **PostgreSQL** – events and metadata
- **hashlib SHA-256** – integrity checks
- **datetime / zoneinfo** – timestamps
- **ReportLab + CSV** – reports

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Optimization Proposed

Six gaps in today's investigations and the CyberTrace feature proposed for each

Gap	Before: problem today	After: proposed CyberTrace feature
1. Evidence collection coverage	Unknown which systems stopped reporting or which time intervals are missing	Source inventory, agent heartbeat, missing-interval records, local buffering, reconcile after reconnect
2. Timestamp uncertainty	Time zones and clock offsets make event order unreliable	Keep original timestamp; convert to UTC only if zone or offset is known; flag unknowns, never guess
3. Evidence integrity and traceability	Hard to tell if evidence changed or where it came from	Originals kept apart from analysis output; SHA-256 at collection; metadata, audit trail, re-verification
4. Explainable correlation	Events linked by hand; reasons unclear	Link by IP, user, host, session or request ID in time windows; show why; mark strong vs possible
5. Transparent parsing failures	Malformed or unsupported records silently shrink the evidence	Count received, parsed, rejected; list missing fields; keep originals; show limits in dashboard and report
6. Reproducible reports	Timeline lacks its sources, methods and limits	PDF and CSV with timeline, hashes, gaps, parse failures, timestamp warnings, link reasons, generation time

Proposed contribution: CyberTrace aims to integrate evidence collection coverage, timestamp uncertainty reporting, evidence integrity verification and explainable cross-system incident reconstruction into one workflow for small teams. These capabilities will be evaluated against existing tools using reproducible synthetic incidents.

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Solution Architecture & Design

Visual: planned flow from authorized log sources to the incident timeline and reports (numbers show the order).

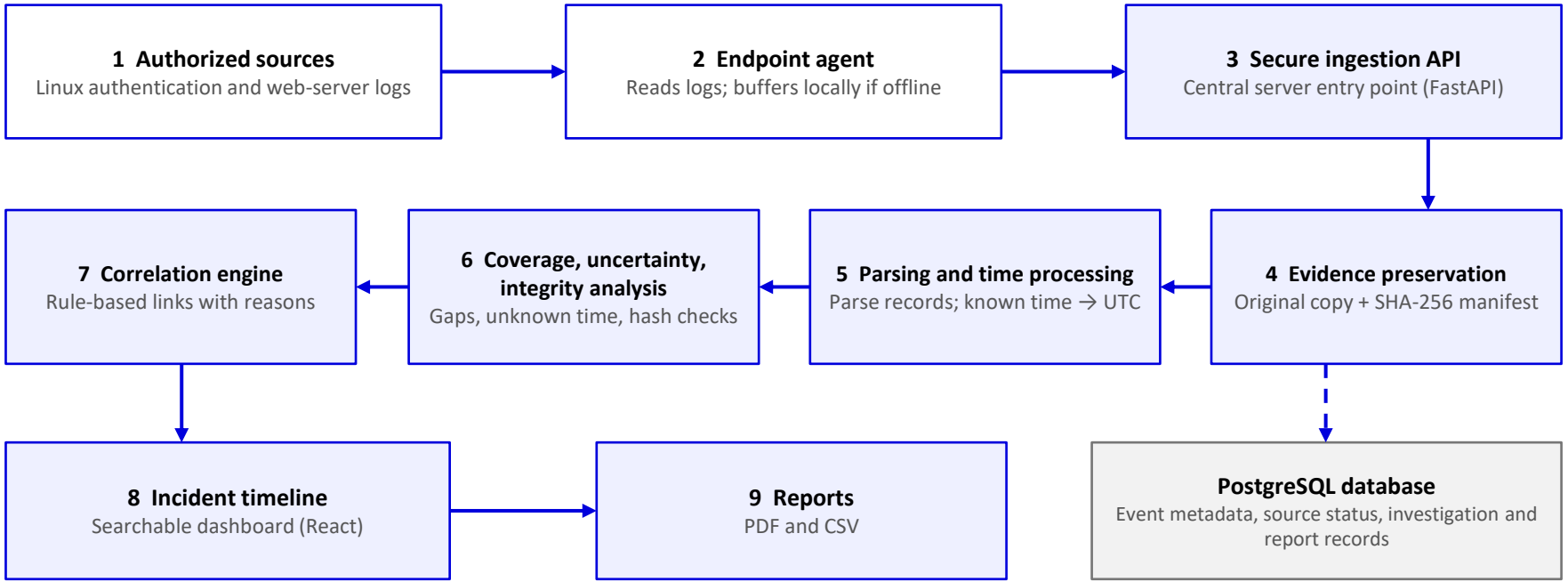


Figure: Planned CyberTrace architecture. Solid arrows show data flow; the dashed arrow shows the database linked to the central server. A hash detects later changes; it does not prove authenticity before collection.

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Data Flow Diagram

Visual: end-to-end workflow, how uncertainty is handled, and the planned test scenarios.



How the workflow handles each situation

Valid timestamp
Normalize to UTC when the source time zone is known; keep the original.

Unknown timestamp or zone
Preserve and flag it. Never invent or silently assume a time zone.

Invalid record
Keep the original; record a parsing failure.

Disconnected source
Report the collection gap; reconcile buffered records after reconnection.

Planned synthetic incident scenarios (repeatable tests)

1 SSH brute force
Repeated failed Linux logins from one source address.

2 Suspicious web requests
Unusual requests in web-server access logs.

3 Multi-system incident
Related events from more than one source, linked with reasons.

Figure: Data flow from collection to report. Validate means checking records, timestamps and hashes before correlation.

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Deployment Timeline

Six build phases, each tested with the synthetic scenarios. Phase dates are still to be set by the team.

1. **Foundation**
   Server, Linux agent, secure ingestion, auth and web logs
2. **Evidence and parsing**
   Original storage, SHA-256 manifest, parsers, failure counters
3. **Time and coverage**
   Timestamp handling; source inventory, heartbeat, buffering
4. **Correlation and timeline**
   Rule-based links with reasons; searchable dashboard
5. **Reports and scenarios**
   PDF and CSV; three synthetic incident scenarios
6. **Evaluation**
   Run tests; compare with existing tools; document results

Planned evaluation measures

- Share of generated test events recovered
- Missing collection intervals identified
- Timestamp accuracy against known test times
- Deliberate evidence changes detected
- Accuracy of event correlations
- Time to produce a usable timeline and report

Expected outcomes

A searchable timeline; visible collection and parsing gaps; verifiable hashes; explicit timestamp uncertainty; explainable links; reproducible PDF/CSV reports; measured results from synthetic scenarios.

Prototype cost: planning estimate ₹0–₹3,000 using open-source tools (hosting and domain optional).

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Bibliography

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6. Plaso (log2timeline) documentation.
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Thank You for Your Attention!